

SAUD AHMAD

AI ENGINEER

saud6446@gmail.com | +92 346 293 6273 | Swat, Pakistan

Portfolio: [saudassist.up.railway.app](#) | GitHub: [github.com/SaudDSxAI](#) | LinkedIn: [linkedin.com/in/saud-ahmad1](#)

PROFILE

AI Engineer and **co-founder** building AI systems that run in real production, not just prototypes. I am currently an AI Engineer at COTER Global and co-founder of **Oval Labs**, an AI automation studio delivering production AI systems for clients in manufacturing, construction, and recruitment. My work spans the full range of AI engineering, including **LLM integration, RAG, agent orchestration, prompt and context design, and model fine-tuning, backed by full stack engineering and cloud deployment.**

EXPERIENCE

Oval Labs, Co-Founder and AI Engineer **Jan 2026 to Present**

Oval Labs is an AI automation studio that turns manual business workflows into systems that run themselves. The studio designs, builds and maintains production software end to end, from architecture and AI design through to live deployment and ongoing support, across recruitment, aftersales, manufacturing and sales operations.

COTER Global, AI Engineer, Contract (Remote)**Sep 2025 to Present**

- Built the hiring pipeline the firm now recruits through, taking it from first requirement to shortlisted candidate without manual handoffs.
- Own the platform in production: ongoing support, iteration and reliability across live requisitions.

[PRODUCTS OF OVAL LABS]

1. Nebula, AI Recruitment Platform

- Recruitment automation platform: hiring requirements parsed by AI into structured roles, job posts generated automatically across 4 platforms, chatbot-based CV intake, and automated candidate screening and shortlisting.
- Automated WhatsApp and Gmail outreach with AI auto-reply, collecting candidate availability at scale without recruiter follow-up.

2. Centauri, AI Aftersales Platform

- Aftersales support platform covering the full ticket lifecycle: AI issue intake that captures customer issues accurately without data entry staff, a customer portal with real-time status tracking and automated approvals, and AI-assisted dispatch that matches the right technician to each job.
- Field companion that lets technicians log diagnoses by video, voice or text, with required fields populated automatically from the input.
- Natural-language analytics over aftersales data, plus a performance dashboard covering resolution times, team performance and customer satisfaction.

3. ZYP Assembly QC System

- Two-gate quality control system: 14-station in-line inspection with operator checklists and verifier sign-off, followed by a separate pre-delivery inspection gate before a bike ships.
- Live analytics dashboard surfaced a 33% first-pass PDI clearance rate that paper tracking had never made visible, giving the plant its first quantified defect signal.

4. Cold Outreach Platform

- Internal client acquisition system: upload a lead list, AI drafts a personalised cold email per lead, delivered under sending rate limits at up to 150 emails a day.
- Automated 3-step follow-up sequence with a different angle at each step, plus reply and bounce detection sorted into a kanban board.

[FREELANCE WORK, 2025]

TrueSight, AI Deepfake Detection

- 3-layer detection pipeline: a video deepfake model (face detection over 40 sampled frames with a 2-factor consensus rule), an audio deepfake model with segment-by-segment timeline scoring, and a rule-based forensic metadata scanner. None of the three depends on the others.
- Pre-flight quality checks that grade whether an uploaded file is good enough to trust before running full analysis.

HSE Performance Tracker (UAE construction firm)

- Multi-tenant safety compliance platform tracking 30 daily HSE fields per worker per project: task briefings, toolbox talks, NCR closures and safety walks.
- Entries roll into a live performance score with monthly KPI tracking and project-level access control, so a site lead only sees their assigned projects. Currently tracking 123 workers across roughly 3,000 man-hours.

PROJECTS

Full writeups, code and results for every project are at [saudassist.up.railway.app/projects](#). Section titles below link straight to each collection.

Agentic AI · 2 projects

Two agents that decide for themselves how much work a question deserves. The first picks its own depth: answer immediately, run a single bounded search, or launch a full research pass that reads five sources live and streams the findings back. The second runs on my portfolio site, choosing turn by turn whether it already knows enough or needs to pull a project's full detail. Both solve the same problem, which is stopping an agent from spending time and money on questions that never needed it.

RAG and Retrieval · 4 projects

Four studies of one question: how do you get a language model to answer from your own documents and actually be right? I built five retrieval approaches and ran them against identical questions side by side, which showed the most sophisticated method was not the best one and documented where two popular techniques broke down. The other three cover what happens next: trimming retrieved text so the context budget stretches further, tying every claim back to the passage it came from, and testing how retrieval quality holds up as the document collection grows.

Fine-Tuning and PEFT · 1 project

Adapting a language model to a new task without paying for a full retrain. I implemented the lightweight adaptation method by hand rather than importing a library, then ran it head to head against a full retrain on the same task, measuring what you actually give up: training time, storage per saved model, and final quality.

Vision-Language and Multimodal · 2 projects

Getting models to connect images and words. One searches a real photo library from a plain English description with no tagging, no training and no labels, relying only on the shared meaning between a picture and a sentence. The other is a small image-captioning system built by hand: a vision model and a language model, both frozen, joined by a translation layer trained from scratch so one can speak to the other.

Deep Learning · 4 projects

Four builds spanning security, image generation, text and audio. The security project finds intrusions in network traffic with no labelled attacks to learn from, reaching 0.95 ROC-AUC across 22,544 connections by learning what normal looks like and flagging what isn't. The generation project builds two competing approaches from scratch and surfaces a real quirk in how one reports its own progress. The other two classify sentiment across 50,000 film reviews, where a simpler baseline won, and recognise short spoken commands from raw audio.

Machine Learning · 6 projects

Six applied problems, each ending in a decision rather than a leaderboard score. Customer churn across 7,032 accounts, where three models were benchmarked and the simplest won on the evidence at 0.836 ROC-AUC. Fraud detection across 283,000 card transactions, where the textbook approach produced a 95% false alarm rate until the evaluation metric was corrected. House prices to 0.924 R² from 216 engineered features, and sales demand forecast six months ahead at 83.3% accuracy. Two are honest negative results: heart disease models that came back statistically tied, and a recommendation engine that only narrowly beat a naive baseline. Both are written up that way.

Robotics and IoT · 6 projects

Six systems that had to work in the physical world rather than on a laptop. A solar-powered field robot that identifies and removes weeds on its own, running vision on-board at 95.6% detection accuracy. A face-recognition entry system deployed at a university campus with 536 logged events, and a second that takes attendance from a live video stream. Alongside these, a home automation controller running entirely on its own hardware with no cloud service behind it, a rover sampling soil moisture to a dashboard, and a robot combining obstacle avoidance, line following and remote control on one chassis.

TECHNICAL SKILLS

LLMs and RAG: OpenAI, Gemini, Claude APIs. LangChain, LangGraph. RAG pipelines (Naive, Hybrid, Re-ranked, HyDE, Agentic), FAISS, sentence-transformers, rank-bm25, cross-encoder re-ranking. Agentic tool-use architectures, SSE streaming.

ML and Deep Learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost. LoRA, QLoRA, GPT-2 fine-tuning, CLIP, YOLOv8, ArcFace, LSTM, DCGAN, U-Net, diffusion models.

Data: pandas, NumPy, SciPy, SHAP, statsmodels (SARIMA), Prophet, matplotlib. Feature engineering, imbalanced-data handling.

Backend and Frontend: Python, C/C++, FastAPI, Next.js, React, WebSocket. PostgreSQL, SQLite, SQLAlchemy, Redis. JWT auth, role-based access control, Tailwind CSS.

Infrastructure: Docker, Railway, Cloudflare R2. Google Sheets, Gmail SMTP/IMAP, WhatsApp (Baileys), LinkedIn and Meta Graph APIs.

Robotics and Embedded: Jetson Orin Nano, Arduino, ESP32, Raspberry Pi. OpenCV, GPS/IMU sensor fusion, MQTT, Bluetooth Serial.

EDUCATION

BSc Data Science, GIK Institute of Engineering Sciences and Technology 2021 to 2025

Coursework: Artificial Intelligence, Data Mining, Natural Language Processing, Big Data Analytics, Data Warehousing and Business Intelligence, Data Visualization, Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Databases, Software Engineering, Parallel Processing, Data and Network Security, Internet of Things, Linear Algebra I and II, Differential Equations, Discrete Structures, Inferential Statistics and Applied Probability.

Technical Lead, Team Techno (Robotics)

Led GIK Institute's robotics team and set technical direction across hardware and software. That team is the foundation behind AgroBot, Access Vision and the other robotics builds here.

CERTIFICATIONS

IBM: Machine Learning with Python | IBM: Python for Data Science and AI | IBM: Data Science Methodology | DeepLearning.AI: Generative AI for Everyone